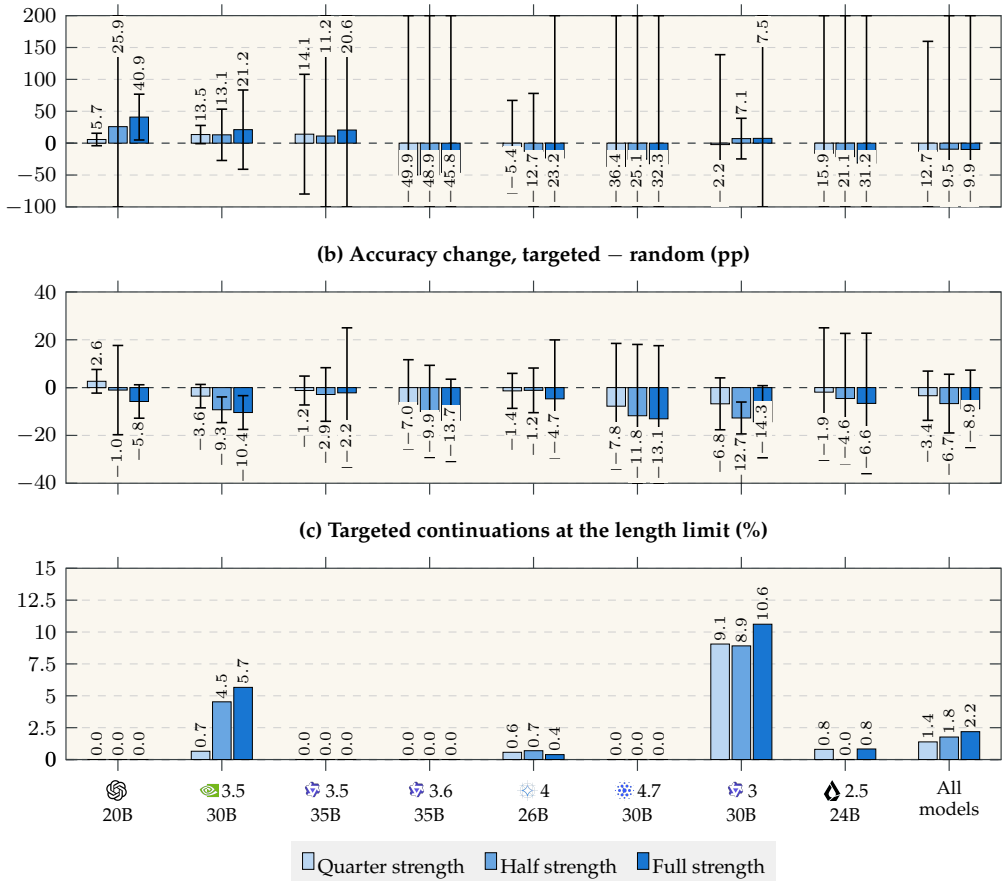

(a) Reasoning-token change, targeted − random (%)
200
150
100
50
0
−50
−100
5.7
25.9
40.9
13.5
13.1
21.2
14.1
11.2
20.6
49.9
48.9
45.8
5.4
12.7
23.2
36.4
25.1
32.3
2.2
7.1
7.5
15.9
21.1
31.2
12.7
9.5
9.9
(b) Accuracy change, targeted − random (pp)
40
20
0
−20
−40
2.6
1.0
5.8
3.6
9.3
10.4
1.2
2.9
2.2
7.0
9.9
13.7
1.4
1.2
4.7
7.8
11.8
13.1
6.8
12.7
14.3
1.9
4.6
6.6
3.4
6.7
8.9
(c) Targeted continuations at the length limit (%)
15
12.5
10
7.5
5
2.5
0
0.0
0.0
0.0
0.7
4.5
5.7
0.0
0.0
0.0
0.0
0.0
0.0
0.6
0.7
0.4
0.0
0.0
0.0
9.1
8.9
10.6
0.8
0.0
0.8
1.4
1.8
2.2
20B
3.5 30B
3.5 35B
3.6 35B
4 26B
4.7 30B
3 30B
2.5 24B
All models
Quarter strength
Half strength
Full strength